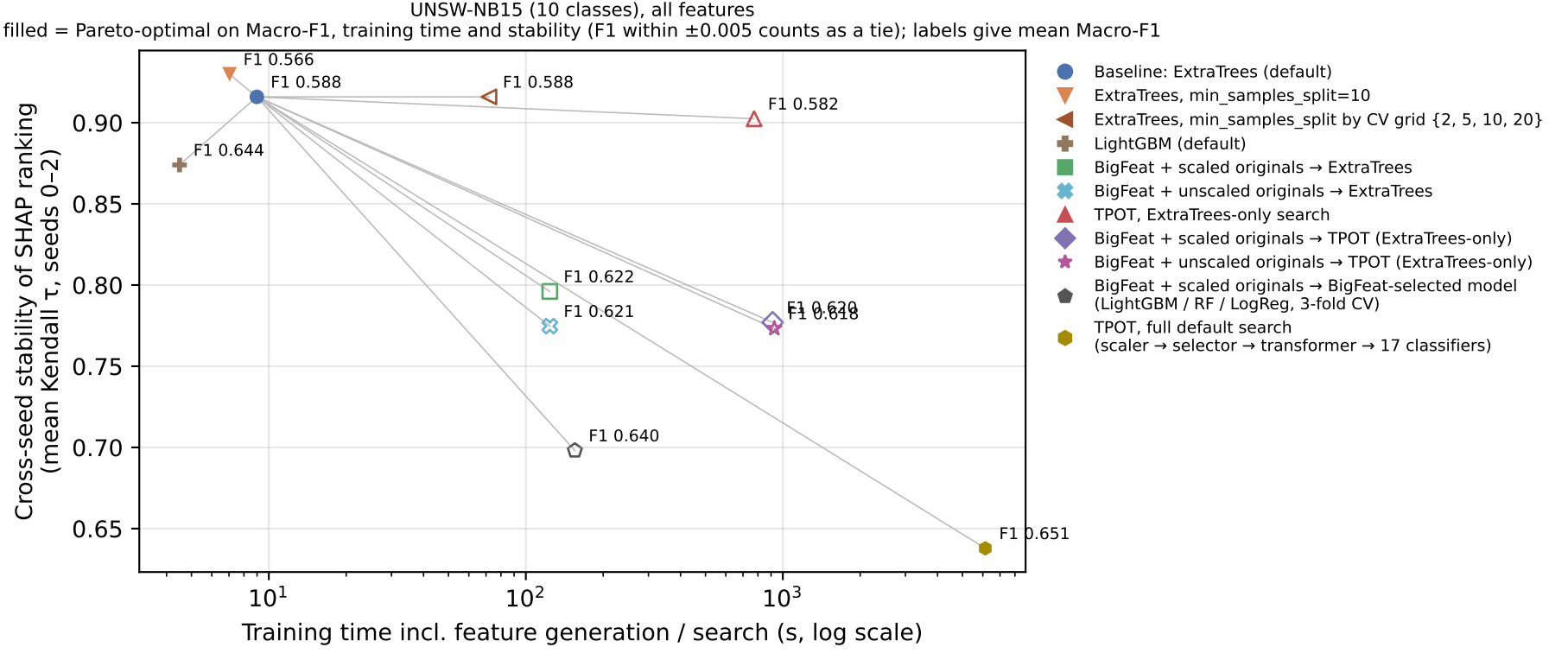
UNSW-NB15 (10 classes), all features
filled = Pareto-optimal on Macro-F1, training time and stability (F1 within ±0.005 counts as a tie); labels give mean Macro-F1
Cross-seed stability of SHAP ranking
(mean Kendall τ, seeds 0–2)
Training time incl. feature generation / search (s, log scale)
0.90
0.85
0.80
0.75
0.70
0.65
10^1
10^2
10^3
F1 0.566
F1 0.588
F1 0.588
F1 0.582
F1 0.644
F1 0.622
F1 0.621
F1 0.620
F1 0.618
F1 0.640
F1 0.651
Baseline: ExtraTrees (default)
ExtraTrees, min_samples_split=10
ExtraTrees, min_samples_split by CV grid {2, 5, 10, 20}
LightGBM (default)
BigFeat + scaled originals → ExtraTrees
BigFeat + unscaled originals → ExtraTrees
TPOT, ExtraTrees-only search
BigFeat + scaled originals → TPOT (ExtraTrees-only)
BigFeat + unscaled originals → TPOT (ExtraTrees-only)
BigFeat + scaled originals → BigFeat-selected model
(LightGBM / RF / LogReg, 3-fold CV)
TPOT, full default search
(scaler → selector → transformer → 17 classifiers)